

Adaptive Region-Aware Geo-Weighted Push-Sum Gossip for Highly Mobile Vehicular Networks

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Abstract—Fleets of vehicles, whether human-driven delivery and ride-hailing fleets or the newer wave of autonomous vehicles (AVs), need to know what is happening around them right now: how fast traffic is moving in their part of the city, without waiting on a central server to tell them. Routing every vehicle's data through one server is the obvious answer, but it comes with obvious costs. The server becomes a single point of failure, a bandwidth bottleneck as the fleet grows, and a place where private location data piles up across jurisdictions. Existing gossip-based averaging protocols solve part of this problem but not all of it: plain gossip ignores geography and wastes bandwidth talking to distant vehicles, while geographic gossip (Dimakis et al.) was built for a static sensor field and only ever computes one number for the whole network rather than separate estimates for different parts of a city. VANET clustering research, on the other hand, is good at organizing vehicles into mobility-aware groups, but it optimizes for how long a cluster survives, not for whether anything computed inside it is accurate. This paper proposes Adaptive Region-Aware Geo-Weighted Push-Sum Gossip (Adaptive-GWG), a leaderless protocol in which vehicles gossip more often with nearby peers than distant ones, use push-sum for the underlying averaging, and let logical aggregation regions merge or split on their own as vehicle density changes, with no vehicle ever acting as a coordinator. We compare Adaptive-GWG against uniform random gossip and a fixed-grid version of geo-weighted gossip across fleets of [100, 500, 1000] vehicles and find it lowers per-region MAPE by [X–Y]% and shortens average geographic hop distance, while keeping message complexity and bandwidth usage close to the fixed-grid baseline. We also look at what this means for autonomous vehicles specifically, since they already broadcast position and speed to nearby vehicles over V2X and could use this kind of local, leaderless aggregation for cooperative perception.

Index Terms—gossip protocols, distributed averaging, push-sum, geographic gossip, consensus, VANET, adaptive clustering, autonomous vehicles, cooperative perception, Intelligent Transportation Systems.

I. INTRODUCTION

A. Motivation and Problem Statement

Picture a ride-hailing or delivery fleet with thousands of vehicles moving across a city, or, increasingly, a fleet of autonomous vehicles trying to plan a safe route. Both need the same basic thing: a live, local estimate of average speed to support congestion warnings, rerouting, and dispatch. The obvious solution is to have every vehicle report to a central server, but this breaks down in familiar ways. If the server goes down, the whole fleet loses its shared picture of traffic. As the fleet grows, the server turns into a bandwidth bottleneck and, eventually, a scalability ceiling. And centralizing raw location traces raises privacy and jurisdictional questions most operators would rather avoid.

So can a fleet of vehicles work this out for itself, using only local, peer-to-peer exchanges, with no central coordinator, and without falling apart when vehicles drop in and out of range or drift between zones?

B. Contributions

This paper makes four contributions, each aimed at a gap the related work in Section II leaves open:

- 1) an adaptive region-management layer that changes regions on its own as vehicle density shifts, merging sparse regions with their nearest neighbor and splitting dense regions into smaller sub-regions, instead of relying on a fixed grid for the whole run;
- 2) inverse-distance-weighted peer selection that biases the gossip step itself toward nearby vehicles, not just something used for routing or for deciding where to store data;
- 3) integrating push-sum aggregation with this adaptive, geo-weighted region management, so that each region's speed estimate stays accurate as vehicles move between regions and as regions themselves split or merge; and
- 4) no leader anywhere in the system, at any level, which removes both the single point of failure and the single point of compromise that come with electing a cluster head.

C. Applicability to Autonomous and Connected Vehicle Fleets

Although we have framed the problem around logistics fleets, it applies just as much to autonomous vehicles (AVs). Current AV stacks already broadcast position, heading, and speed to nearby vehicles through V2X messages such as

cooperative awareness or basic safety messages, so the peer-to-peer link Adaptive Region-Aware Geo-Weighted Push-Sum Gossip (Adaptive-GWG) needs already exists. Cooperative perception, where nearby AVs share what they sense to build a fuller picture than any single vehicle's sensors, is essentially the same problem as per-zone aggregation, with the same churn as vehicles enter and leave a neighborhood at highway speed.

To be clear about scope, Adaptive-GWG is meant as a decision-support layer for routing and congestion detection, not a replacement for the real-time collision-avoidance messaging that dedicated V2X safety layers handle; push-sum's eventual-consistency guarantee suits decision support but not the deterministic latency that safety-critical messaging needs.

II. RELATED WORK

A. Gossip-Based Aggregation and Consensus

Gossip-based aggregation goes back to the epidemic replication protocols of Demers et al. [1], sharpened later by Karp et al.'s push-pull analysis [2]. Kempe, Kleinberg, and Demers [4] were among the first to bias peer selection by distance, showing that a power-law decay could keep convergence nearly as fast as uniform gossip while using far less long-distance bandwidth. Kempe, Dobra, and Gehrke [7] then extended gossip from spreading rumors to computing aggregates through push-sum: each node keeps a (value, weight) pair, splits it with a partner each round, and because the network-wide sum never changes, every node's estimate provably converges to the true average. How fast this happens depends on the spectral gap of the gossip matrix [7], [11], [15], a quantity that keeps showing up across this whole literature.

The closest piece of prior work to what we are proposing is Dimakis, Sarwate, and Wainwright's Geographic Gossip [17]. The problem they set out to solve is that plain gossip wastes almost all of its communication budget on partners that are already close together in the network graph, so a flat sensor field can need close to $\Theta(n)$ rounds before every node's running average is even roughly correct. Their fix combines two ideas that had previously been used separately: geographic routing, to reach a partner far away in only a few hops, and push-sum averaging, to combine values once that distant partner is found. Each node routes a message, hop by hop, toward a randomly selected geographic target and only gossips locally, with a nearby node, once the message arrives there. On a random geometric graph this cuts the number of rounds needed for ϵ -accuracy from $\Theta(n)$ down to $O(\sqrt{n} \log n)$, and it is one of the earliest protocols to combine geographic awareness with averaging in a single mechanism. But it was only tested on static sensor fields, it computes one global average rather than separate per-region estimates, and it has no concept of a region that can grow, shrink, split, or merge. Our proposed method, Adaptive Region-Aware Geo-Weighted Push-Sum Gossip (Adaptive-GWG), picks

partners differently: instead of routing toward a randomly chosen target and then gossiping locally, it performs inverse-distance-weighted peer selection directly among the vehicles already reachable within a region, so closer peers are contacted more often without any routing step. We also replace their single flat average with per-region push-sum tracks, so dense and sparse parts of a city can each converge on their own terms instead of being folded into one city-wide number.

Closer to our own setting, Wei et al. look at gossip inside a VANET rather than a static sensor field or a small multi-agent testbed [35]. Their concern sits one layer below ours: before a vehicle can gossip with anyone, it first has to work out who its current neighbors are, and this is harder in a VANET than in a fixed sensor deployment because the neighbor set changes every few seconds as vehicles overtake, turn, or drop out of range. Their GSIM-ND algorithm disseminates "who is here" information through gossip, adds sensing information supplied by roadside units, and takes advantage of multi-packet reception, which lets a receiver decode several overlapping transmissions at once instead of treating them as a collision. Simulation results show this converges 40–90% faster than purely random, scan-based, or plain gossip discovery. This complements our work rather than competing with it: Wei et al. solve the problem of finding a gossip partner in a fast-moving network in the first place, while Adaptive-GWG assumes that problem is already solved and focuses on what happens once a partner is found, namely how the shared value, a regional speed estimate in our case, is combined and how the region itself grows or shrinks. Their protocol also depends on roadside-unit coverage and stops once neighbors have been found; it does not aggregate any shared quantity and has no notion of a region at all, so the two lines of work sit at different layers of the same system and could plausibly be combined in a future deployment, with GSIM-ND handling partner discovery and Adaptive-GWG handling what is done with each partner once found.

Broadcast Gossip [20] and the consensus literature [10], [14], [16] generalize the same averaging idea to broadcast channels and to directed, time-varying topologies, which gives a connectivity argument (joint connectivity over time) that fits vehicular mobility reasonably well, but almost all of this work is tested on small multi-agent testbeds, not at the scale or density of real vehicle traffic.

B. VANET Clustering and Region Management

VANET clustering borrows its basic vocabulary from general MANET routing, particularly Jiang et al.'s CBRP [26], where an elected cluster head handles membership and routes between clusters. Mukhtaruzzaman and Atiquzzaman's survey [28] groups the VANET clustering that followed into intelligence-based, mobility-based, and multi-hop families. All of them, though, are judged on the same kind of metric: how long a cluster head lasts, how often clusters need to re-form, how much control overhead this costs, not on whether anything computed inside the cluster is actually accurate.

Two region-based, density-aware protocols come closest to what we mean by adaptive regions. SDPC [29] predicts where a vehicle is likely to be a few seconds from now, including whether it is about to go through an intersection, and uses that to pick cluster heads likely to stay stable the longest. RCMS [30] instead models connectivity and overlap between clusters together, grouping vehicles to minimize redundant coverage while keeping the network connected. Both adjust to density and mobility in real time, which is exactly the kind of behavior we need, but neither one gossips or aggregates anything, neither reasons about convergence the way the gossip literature above does, and neither has an explicit split/merge rule for regions that is not just a re-run of cluster-head election.

C. Geographic Routing, Geocast, and Edge Infrastructure

Greedy Perimeter Stateless Routing [34] and geographic hash tables [5] give us the basic tools for addressing a location or a region, but both are built to move a single message to a destination, not to refine a running estimate through repeated exchange. Edge, fog, and blockchain proposals for the Internet of Vehicles [32], [33] solve a different problem entirely: trust, offloading, monetization. These are useful concerns, but mostly orthogonal to whether an aggregate computed across a region of vehicles is accurate, and on-chain consensus is, in any case, far too slow for the sub-second budget a vehicular protocol has to work within.

D. Research Gap

Table I lines up the families discussed above against the properties Adaptive-GWG actually needs: no leader, tolerance for mobility, geographic awareness, regions that adapt, and a real averaging guarantee.

TABLE I. COMPARATIVE POSITIONING OF RELATED WORK

Family	Leaderless	Mobility Tol.	Geo-Aware	Adaptive Reg.	Avg. Guarantee
Push-Sum / Consensus [7],[10],[11]	Yes	Limited	No	No	Yes
Geographic Gossip [17]	Yes	No (static)	Yes	No	Yes
VANET Gossip / Neighbor Discovery [35]	Yes	Yes	Partial (RSU-assisted)	No	No
VANET Clustering [26],[28]	No	Yes	Partial	No	No
Region-Based Adaptive (SDPC/RCMS) [29],[30]	No	Yes	Yes	Yes	No
Adaptive-GWG (Proposed)	Yes	Yes	Yes	Yes	Yes

Push-sum and its relatives have the averaging guarantee but no adaptive regions. VANET-specific gossip work is mobility-tolerant and partly geo-aware through roadside sensing, but stops at neighbor discovery and never aggregates anything. Region-based VANET clustering has adaptive, mobility-tolerant regions but no averaging guarantee. Nothing we found has all five properties at once.

This picture also matches the wider VANET dissemination literature. Shahwani et al.'s recent survey of data dissemination techniques in VANET organizes existing push- and pull-based schemes, including several gossip and epidemic-style methods, mainly around delay, reliability, and overhead trade-offs for delivering a message, not around producing a statistically accurate aggregate [36]. None of the schemes it catalogs update a running estimate the way push-sum does, and none of them treat region formation as something that adapts to vehicle density. This is a useful outside check on our own reading of the literature: the gap we are pointing to is not an artifact of which papers we happened to compare in Table I, but a genuine split between how the dissemination and mobility-management communities usually frame the problem.

This is a gap in how these pieces fit together, not a missing piece on its own; every individual component above already exists somewhere in the literature. What is missing is a

protocol that treats density-driven regional decomposition as part of the aggregation mechanism's own convergence story, rather than treating regions as something handed to it externally by a clustering algorithm, or ignoring regions altogether and treating the whole network as one graph. That is the gap Adaptive-GWG, described in Section III, is meant to close.

III. SYSTEM MODEL

[To be completed — network model, notation, and formal per-region aggregation problem statement, drawing on the system architecture already documented in the Assignment 2 report.]

IV. PROPOSED METHOD: ADAPTIVE-GWG

[To be completed — peer-selection rule, push-sum exchange, and density-triggered region split/merge mechanism, re-leveled from Assignment 2 Sections 1–6 into research-paper tone.]

V. EXPERIMENTAL SETUP

[To be completed after the real NYC Yellow Taxi (2013) dataset is loaded and the failure/churn and sensitivity-

analysis scenarios are run — see project to-do list. Do not populate this section with synthetic-only figures.]

VI. RESULTS AND DISCUSSION

[To be completed — convergence speed, per-region MAPE, hop distance, bandwidth, with confidence intervals rather than $\pm$ std alone.]

VII. LIMITATIONS AND THREATS TO VALIDITY

[To be completed — simulation-vs-deployment gap, scalability assumptions S1–S5 reframed as limitations, AV safety-layer scoping caveat from Section I-C.]

VIII. CONCLUSION

[To be completed.]

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